

Week 3 Progress Meeting presentation

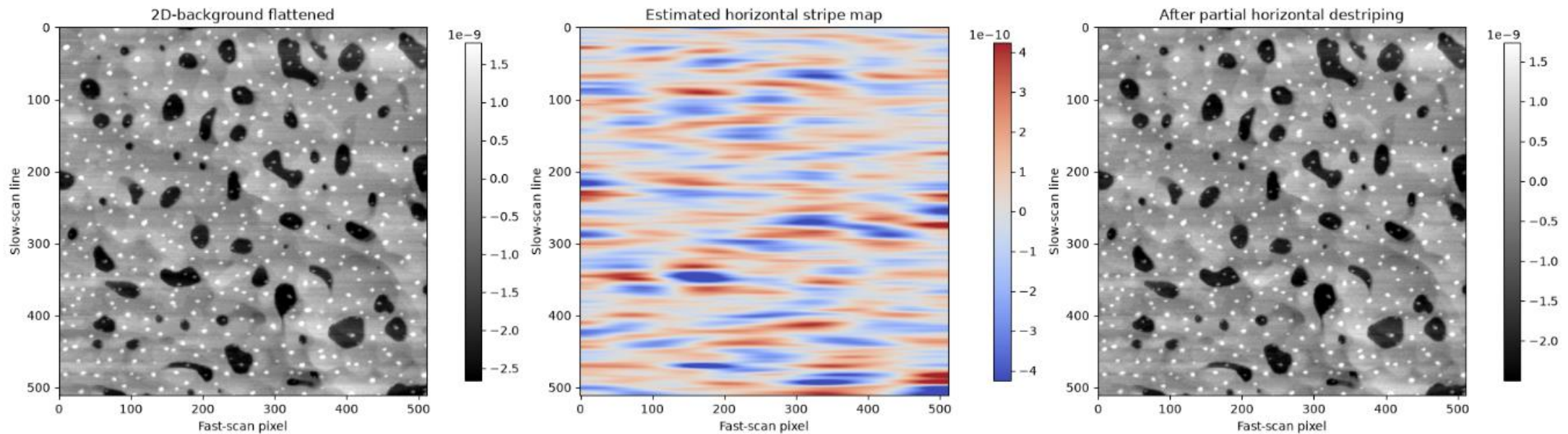
By Poline S

Main changes to the code

- Added preprocessing techniques into the QD detection GUI. And a way to choose algorithm used for QD detection.
- Working on the ground truth dataset:
 - Obtained the data, now in progress of manually labelling QDs – 3 sets ready
 - Segmentation model code ready to run
- Developed a method to delete horizontal stripes manually and in a custom way so that Gabor wavelets and Phase congruency methods are able to work
- Looked at the distributions of QDs within and outside the GaN layer
- Now the algorithm is able to detect and count dots within the GaN layers

Custom pre-processing – Smoothing with different standard deviations along x and y

Quantifying better preprocessing.
dots, pits, average the remainder a
draph to find the st dev of the deviat

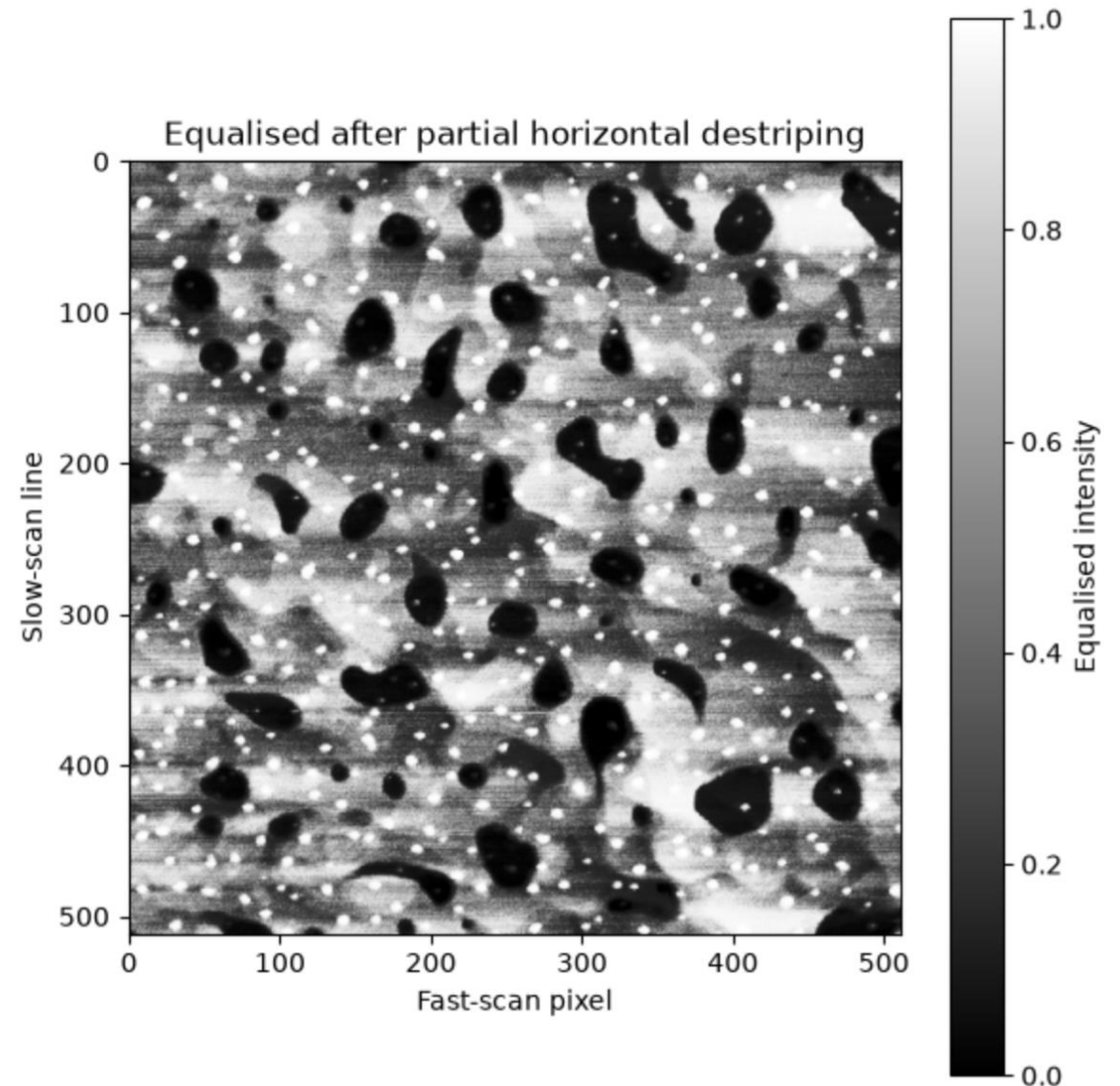


Flat
with
data

I have then compared the previous transformations across the striped and destriped images, and seemed like destriped ones were working better! And images didn't change significantly, judging by eye..

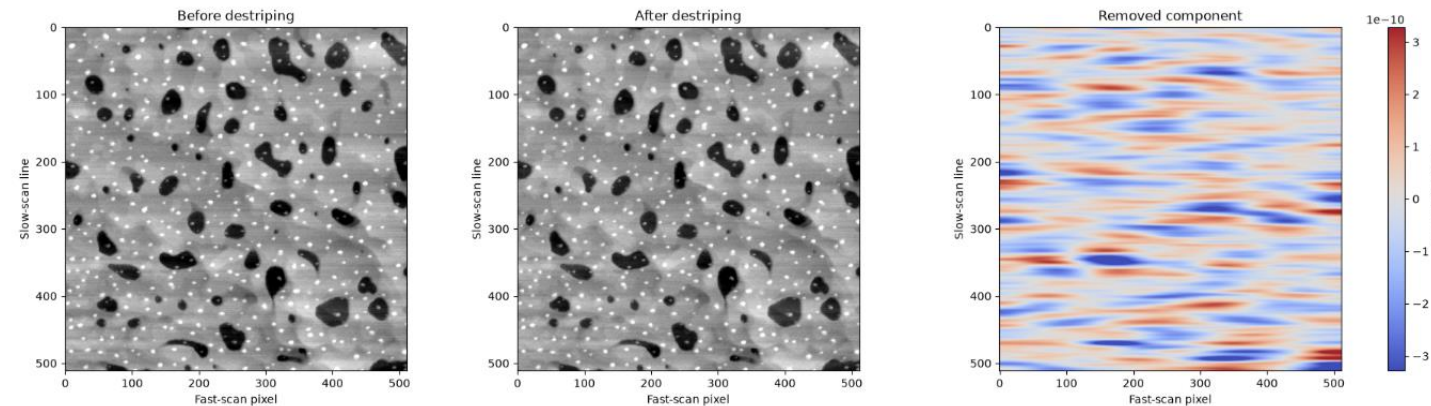
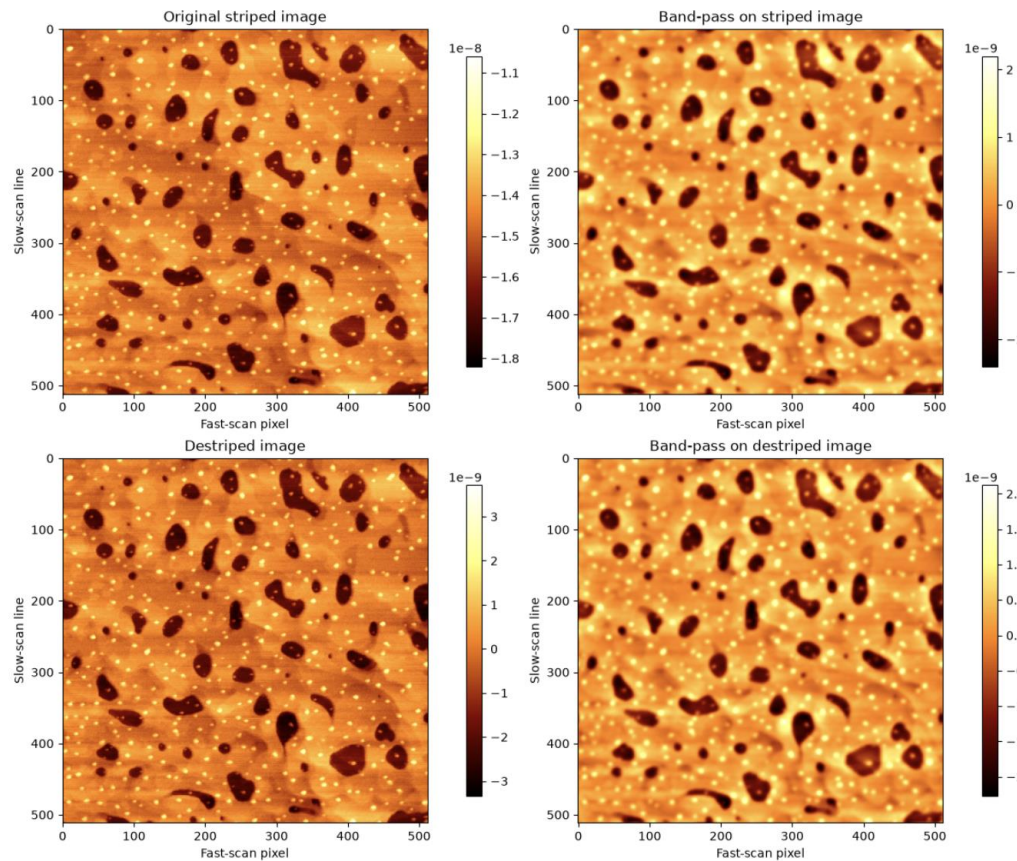
Image equalisation

- Tried redistributing the image's histogram, so that the pixel brightness changes across a higher range of values
- However, this made the horizontal stripes even more apparent, so decided not to follow through this idea



FFT on striped and destriped AFM scans

Test: calculating the removed component and the correlation between images



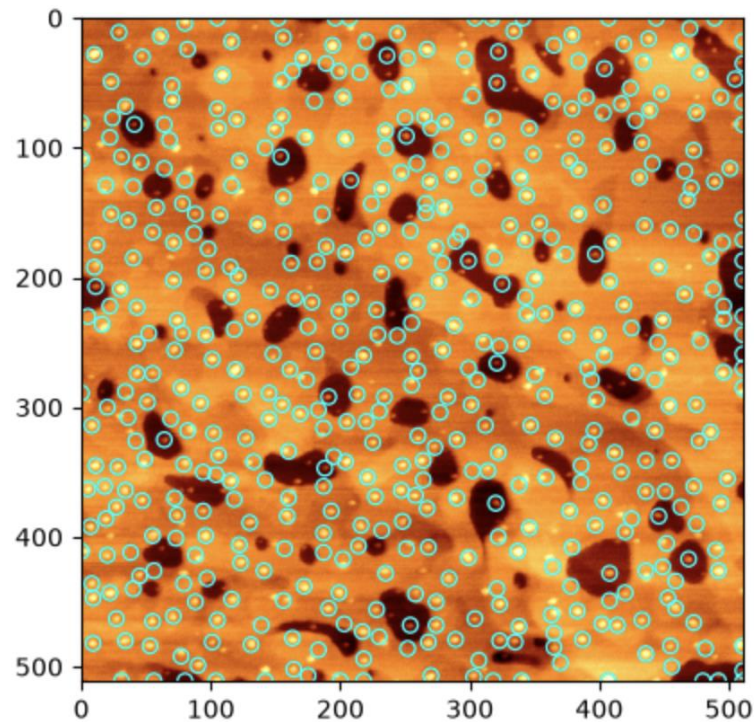
Before/after correlation: 0.9924

Apply to raw data

Relative raw-image change: 0.1918

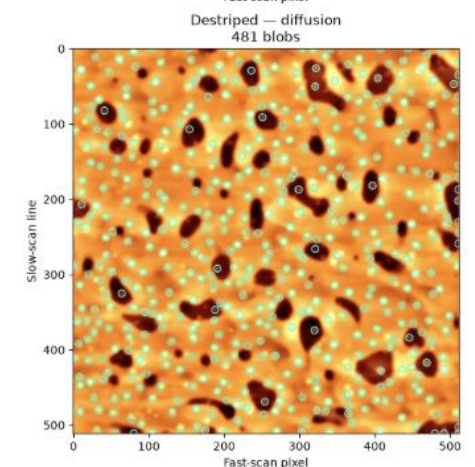
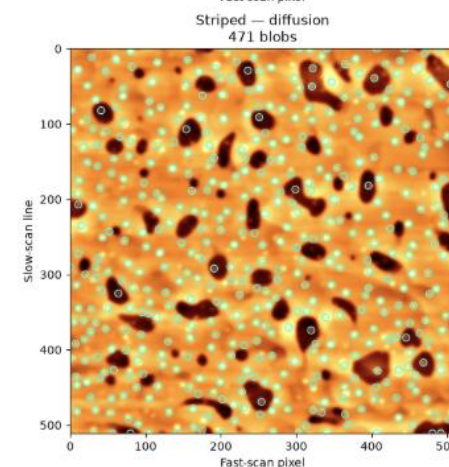
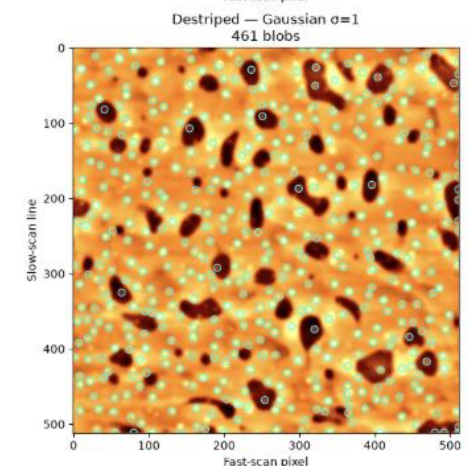
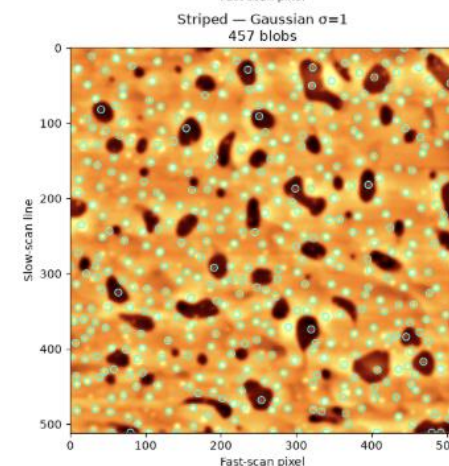
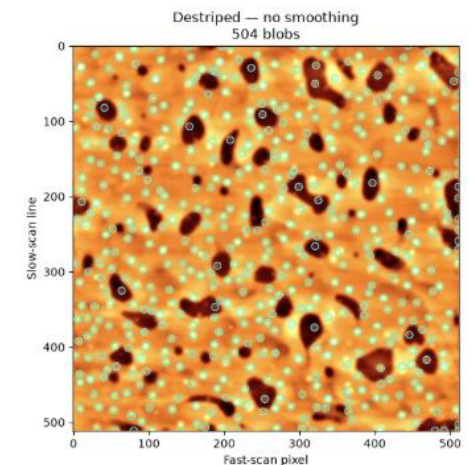
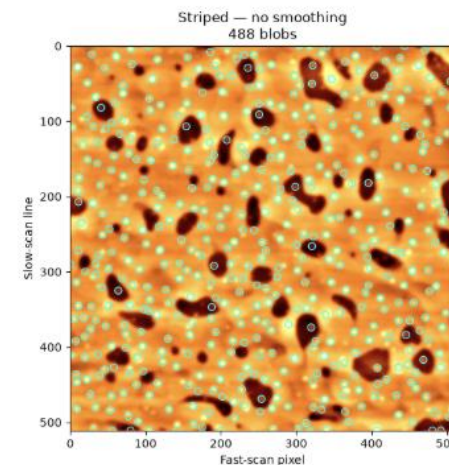
Relative band-pass change: 0.1509

Rerunning previous methods of blob detection on the Destriped image

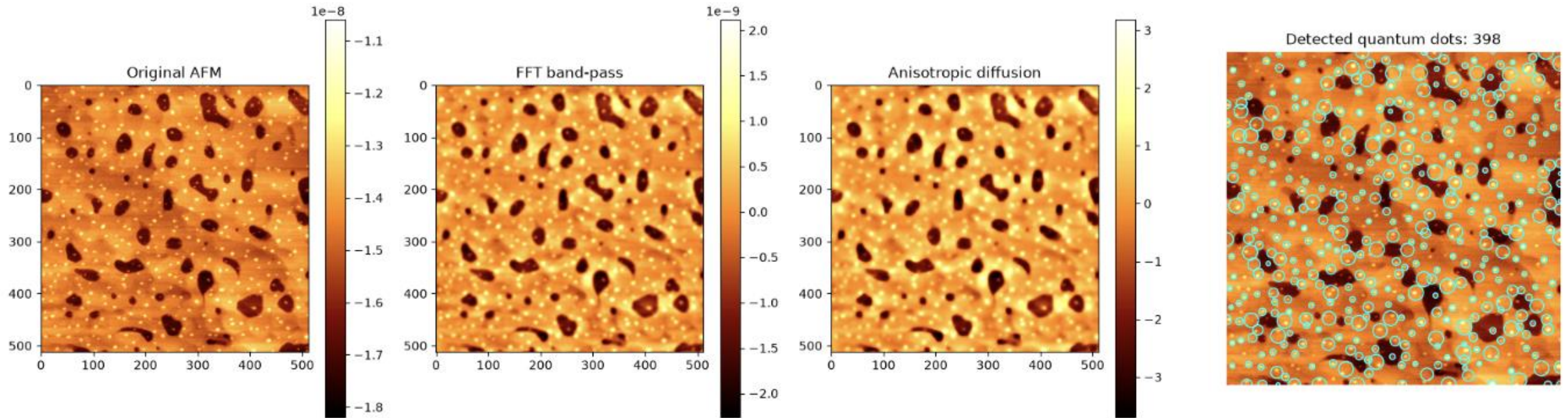


LoG;
Otsu Thresholding remained very similar to before

Add false positive and false negative numbers



Anisotropic Diffusion (also working better)



Mean absolute change: 0.06560607574514782

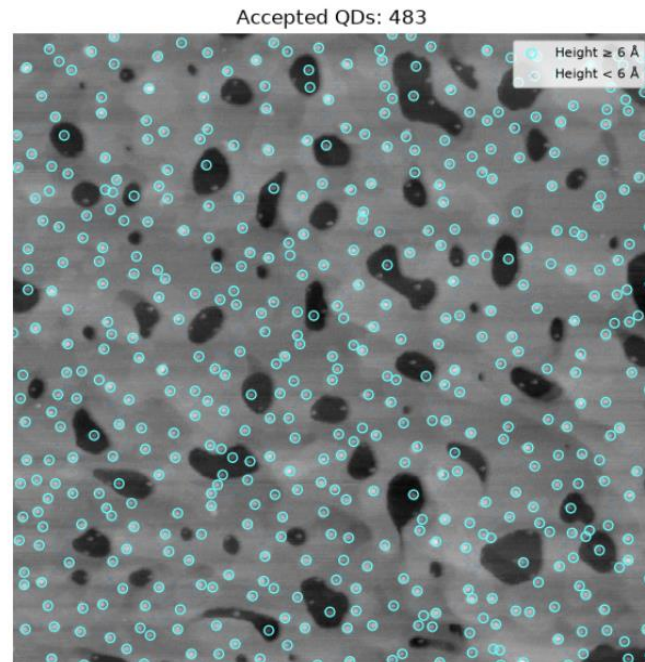
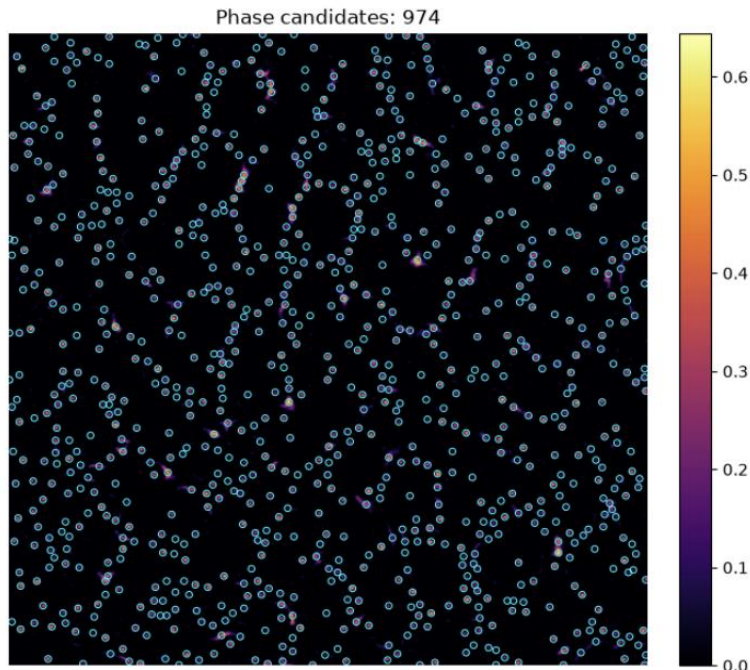
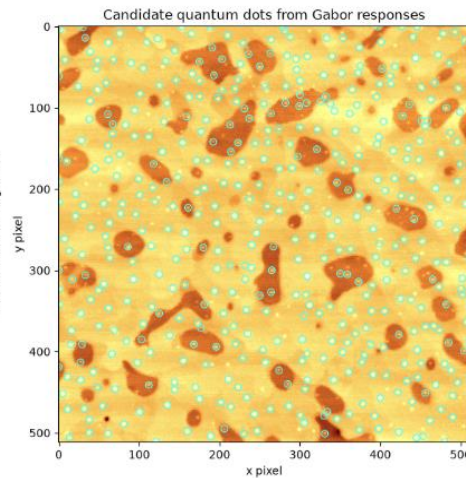
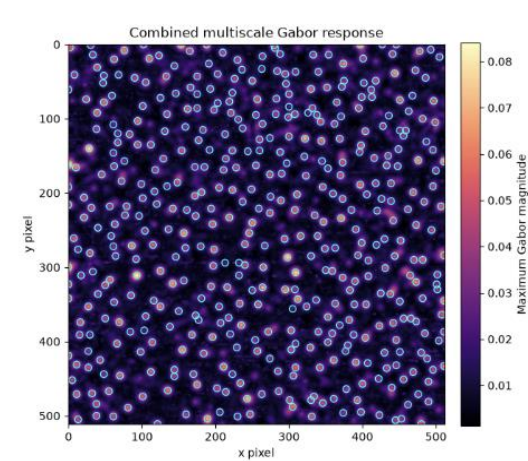
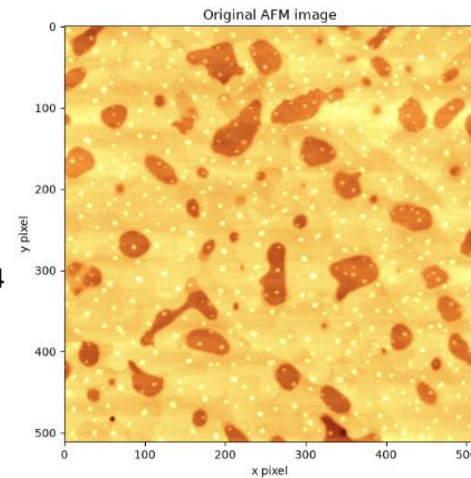
Maximum

absolute change: 0.5586258536273596

Correlation: 0.9960987258391678

Phase Congruency results improved too

Peaks before height filtering: 974
Candidate height range: -0.014921588055659005 0.2545944376921264
Phase candidates: 974
QDs at least 6 Å high: 483
Minimum accepted height: $6.0438950000000001e-10$
Median accepted height: $1.9245329999999999e-09$
Maximum accepted height: $3.3675894999999999e-09$

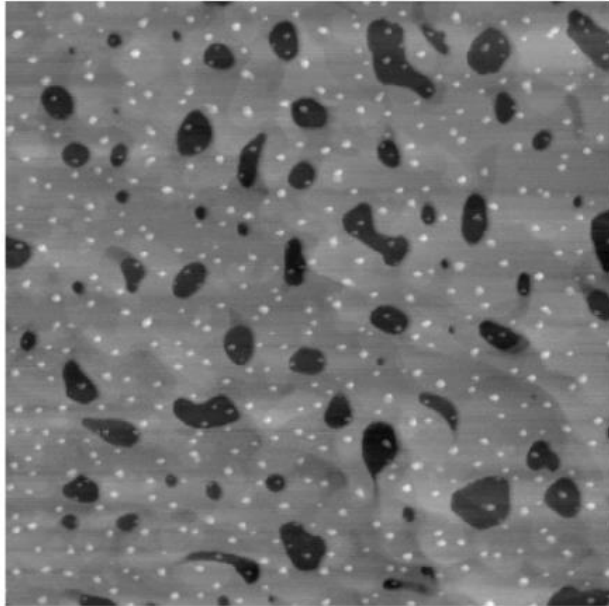


Gabor improved too! But this was expected, since Phase Congruency is based on Gabor.

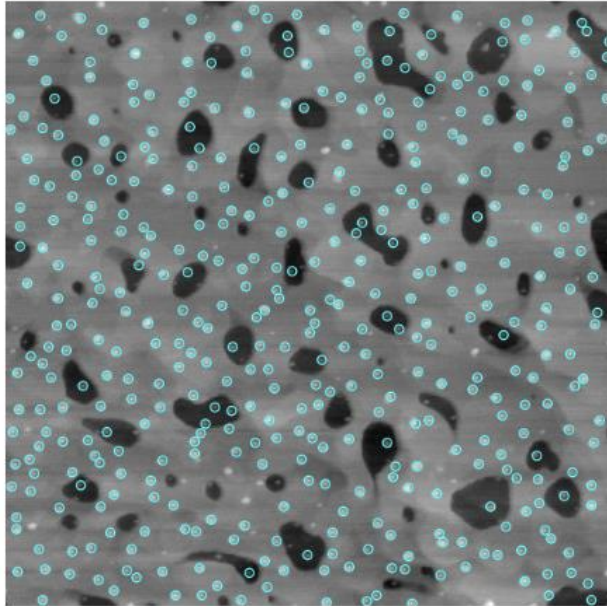
- 75% of pixels have Gabor magnitude less than or equal to 0.012
- 90% of pixels have magnitude less than or equal to 0.021
- 95% are less than or equal to 0.030
- 99% are less than or equal to 0.055

SIFT invariance working too now

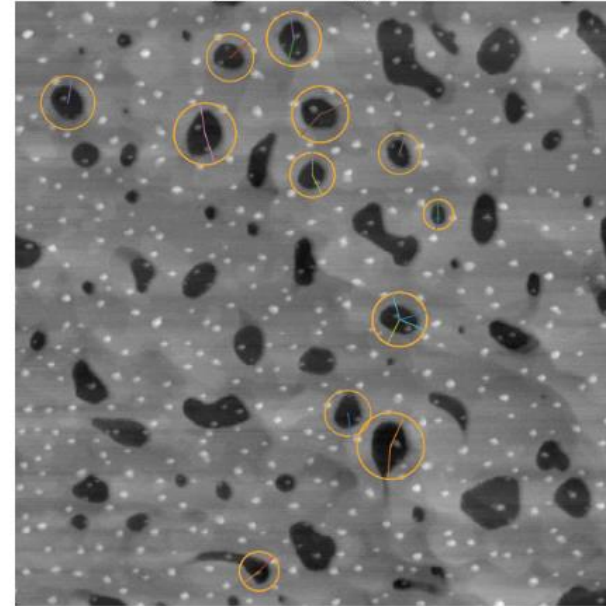
Original AFM image



SIFT QDs $\geq 6 \text{ \AA}$: 426



Large-scale GaN keypoints: 22



SIFT selects points where the Difference-of-Gaussians response forms a strong, well-localized extremum. A very large bright region may be missed because:

- its interior is fairly uniform;
- its boundary is broad rather than sharply localized;
- it is irregular or elongated;
- its strongest responses occur at smaller internal structures;
- $c_edge=20$ may still reject or split parts of it depending on local curvature;
- the available scale space may not match its actual width.

	keypoint_id	y	x	sigma_pixels	orientation_radians	orientation_degrees
0	0	179.908157	355.778687	5.034929	1.716172	98.329405
1	1	481.672974	204.962921	6.655426	2.493008	142.838831
2	2	31.683620	234.139725	9.156921	1.813650	103.914498
3	3	47.583298	180.429306	7.942227	-0.748709	-42.897864
4	4	85.639824	43.928291	9.051963	-1.423358	-81.552425
5	5	95.353645	256.258026	9.520630	2.303198	131.963530

Multiple Otsu for detecting multiple Object classes

Multi-Otsu does not know what a GaN, substrate, or QD is. It only separates the histogram into three groups.

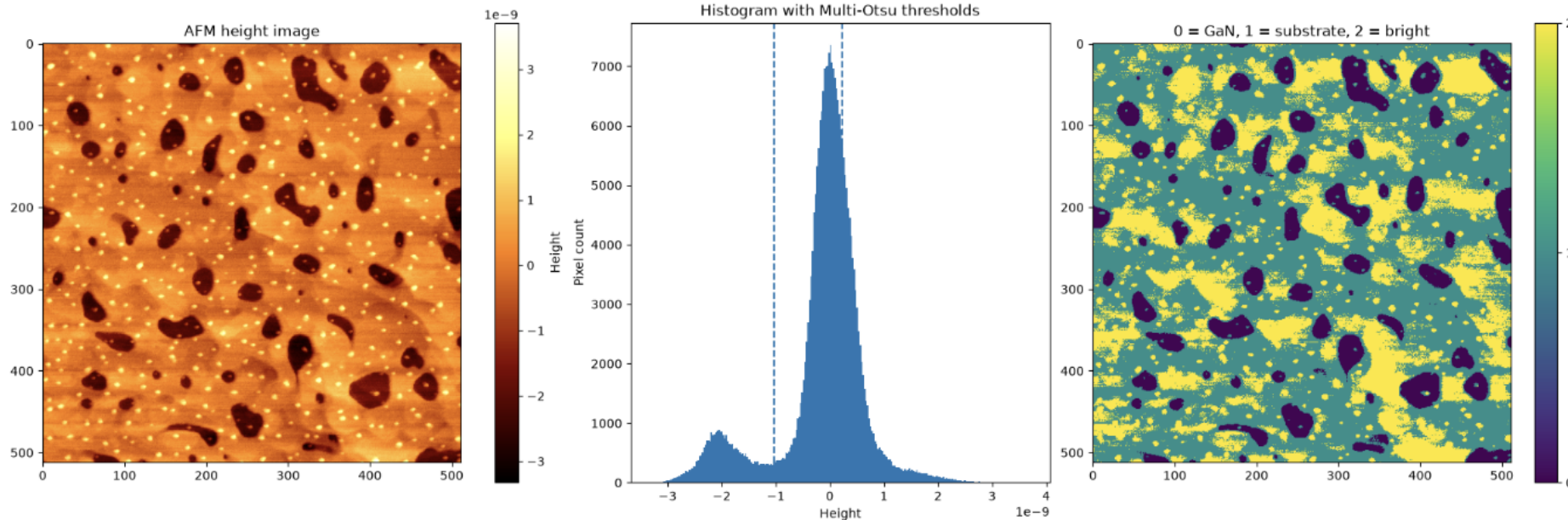
- 0 = GaN
- 1 = InGaN
- 2 = bright QD

Multi-Otsu thresholds: $[-1.03245257e-09 \quad 2.32274196e-10]$

GaN-class pixels: 30926

Background-class pixels: 156034

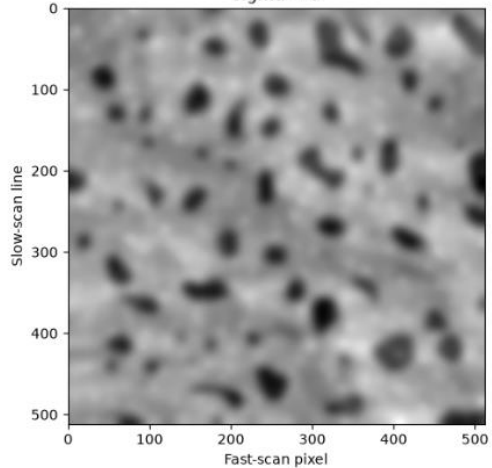
Bright-class pixels: 75184



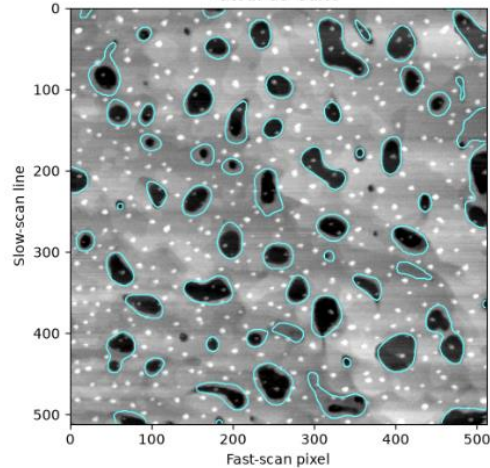
GaN layer detection

Comparing different thresholding methods:

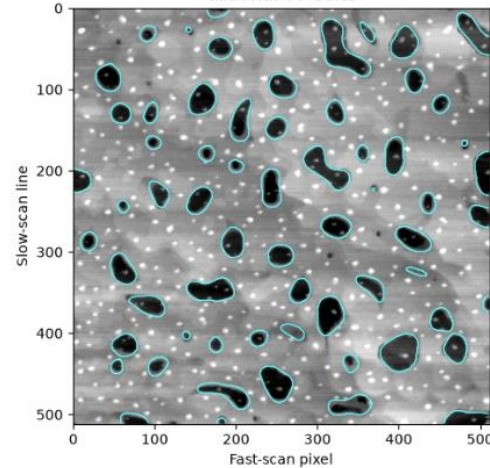
Gaussian-smoothed image
sigma = 6



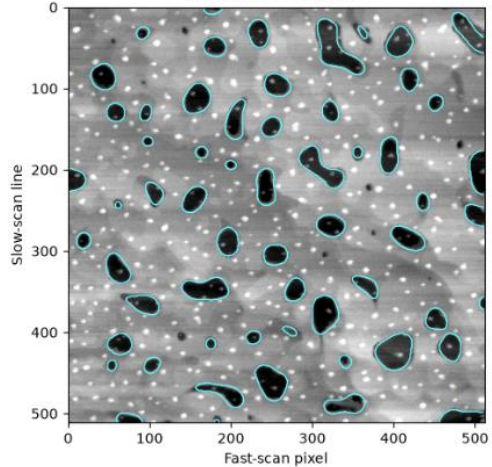
Otsu: 63 GaNs



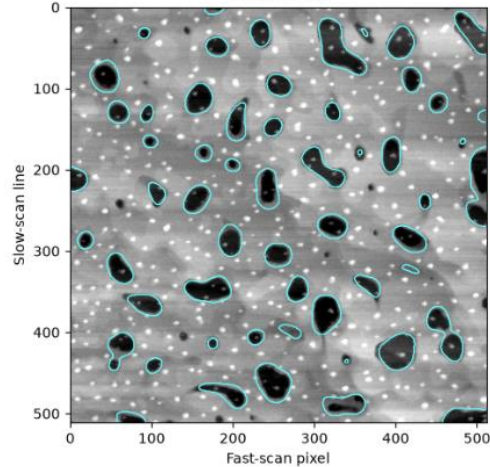
Sauvola: 65 GaNs



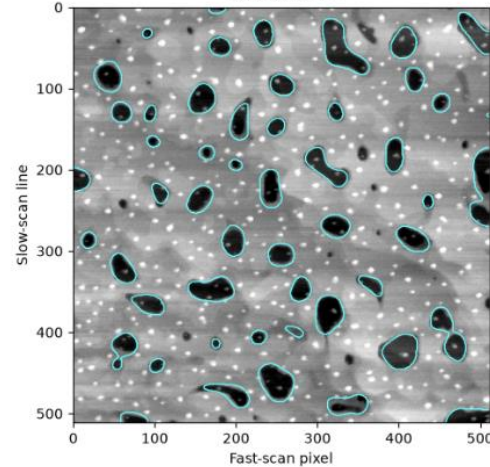
Phansalkar: 60 GaNs



Li: 57 GaNs

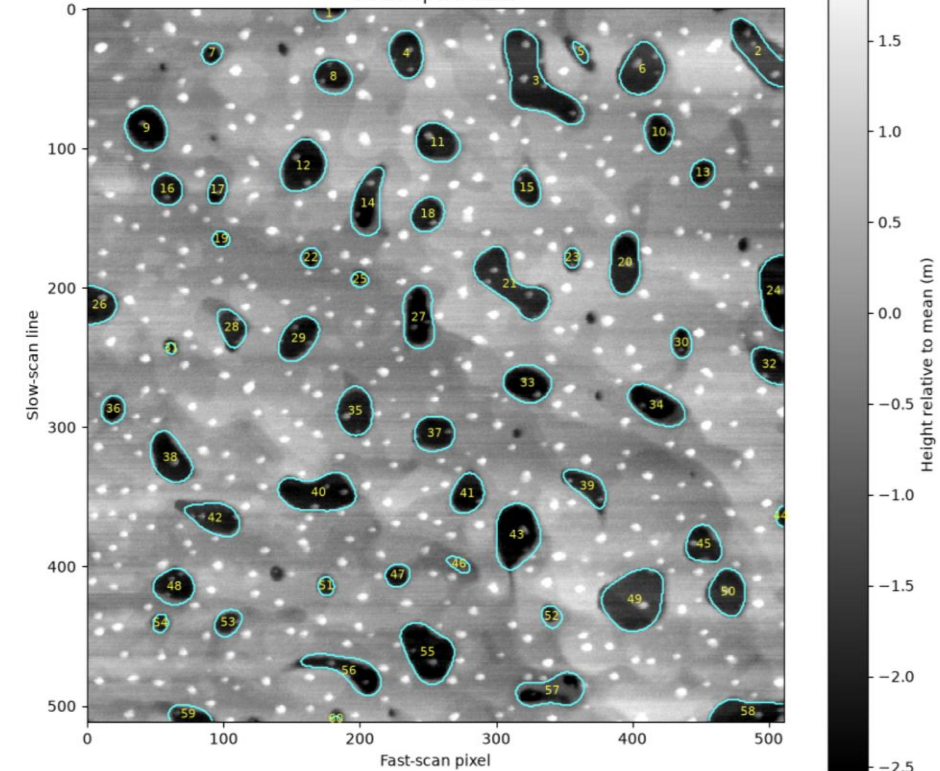


Yen: 52 GaNs

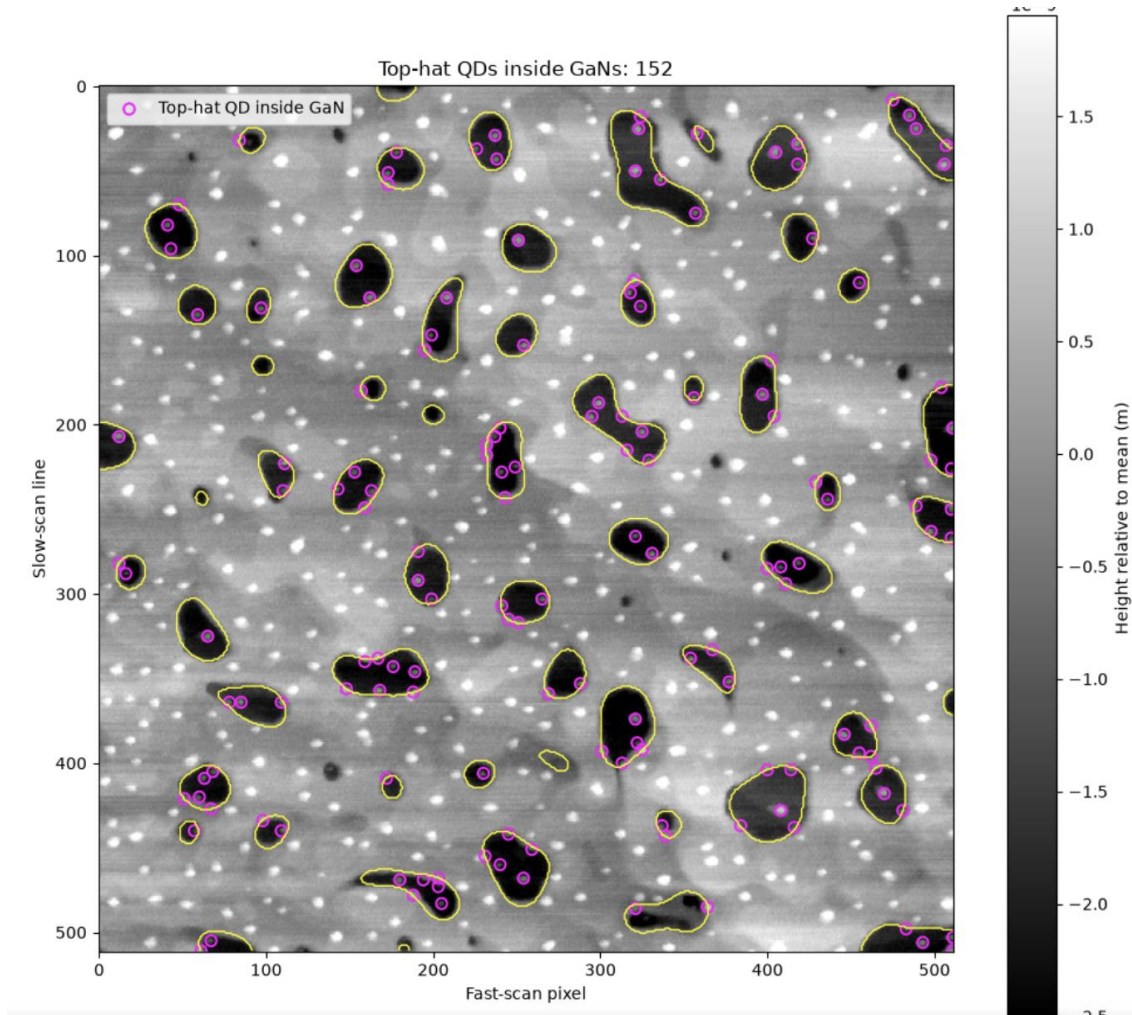


Best Chosen method:

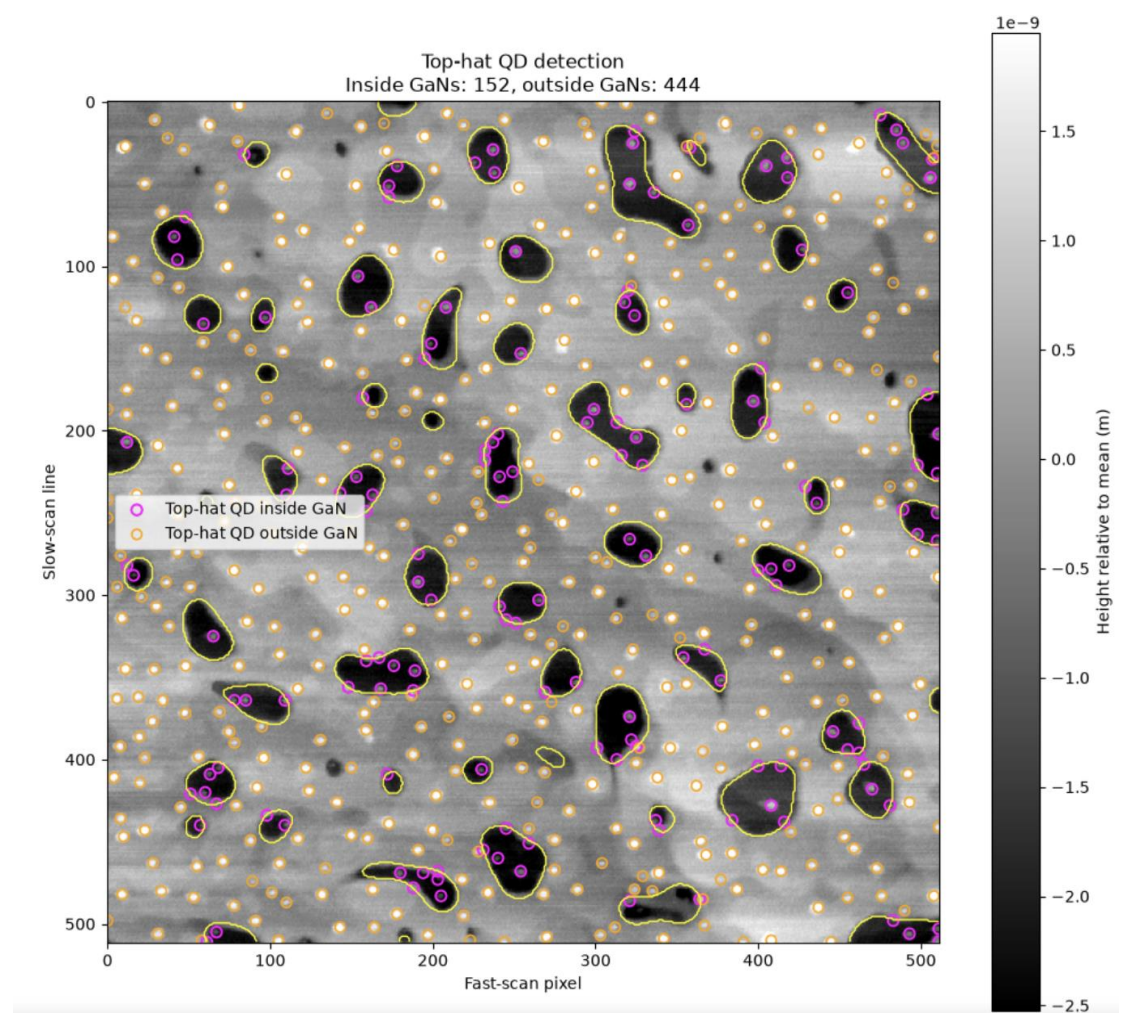
Phansalkar GaN detection
60 accepted GaNs



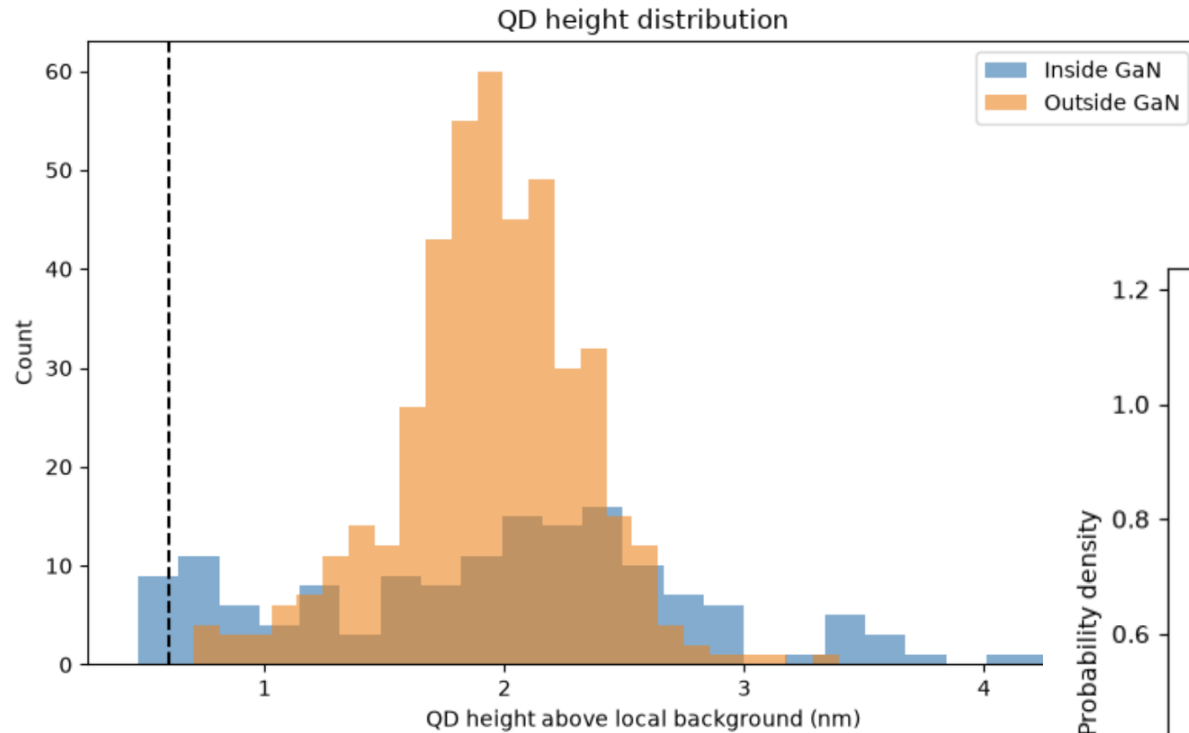
Detecting QDs within GaN layer only



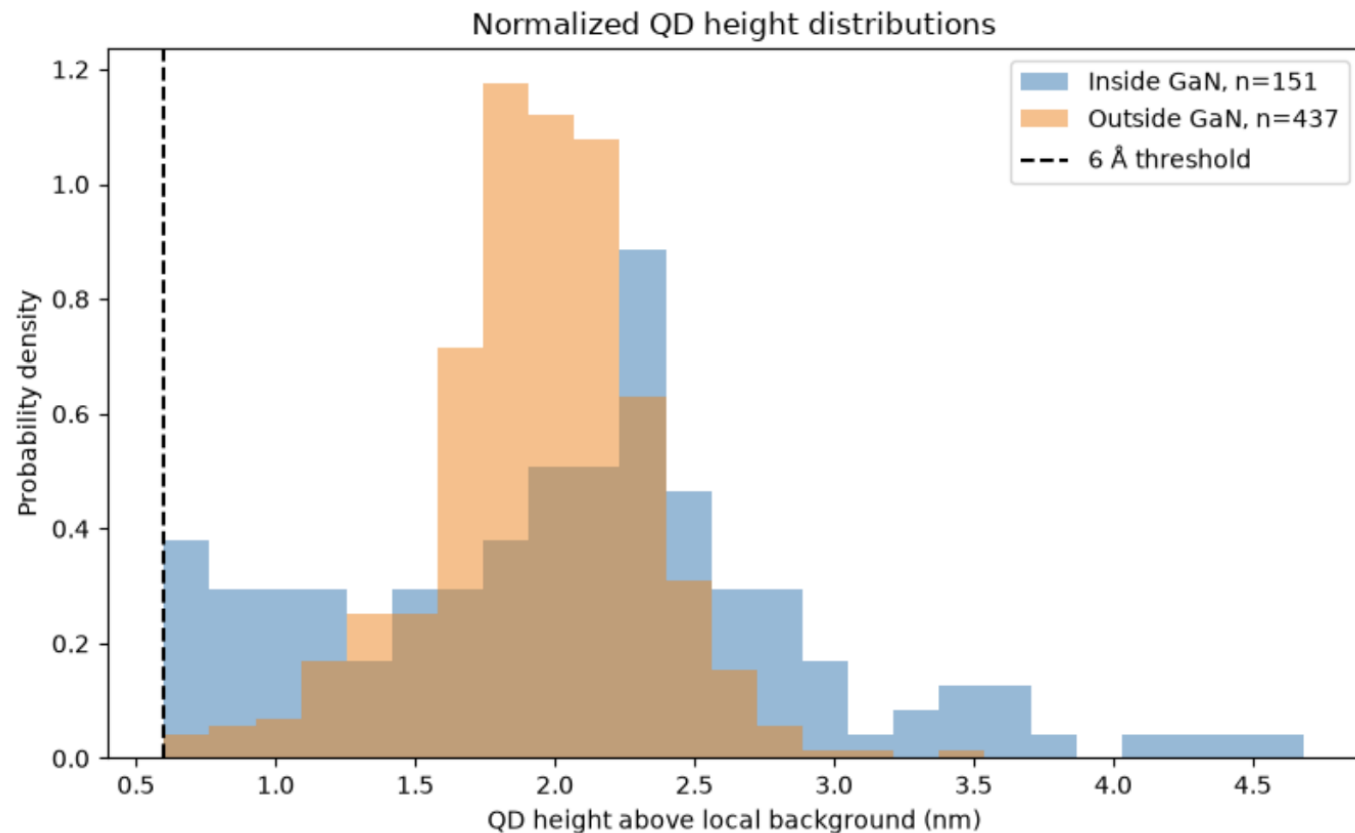
Detecting all possible QDs



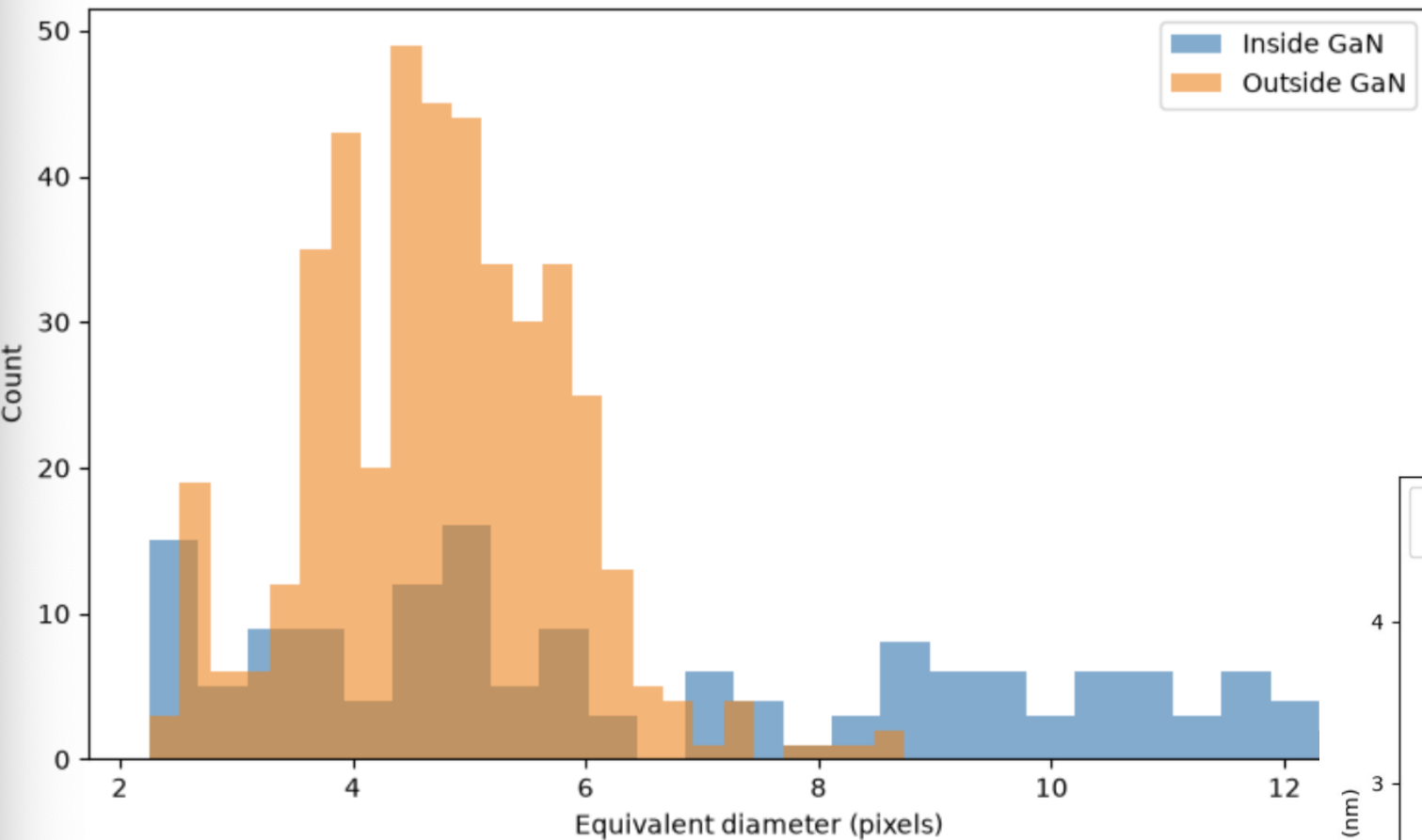
Comparing QDs on GaN and InGaN layers



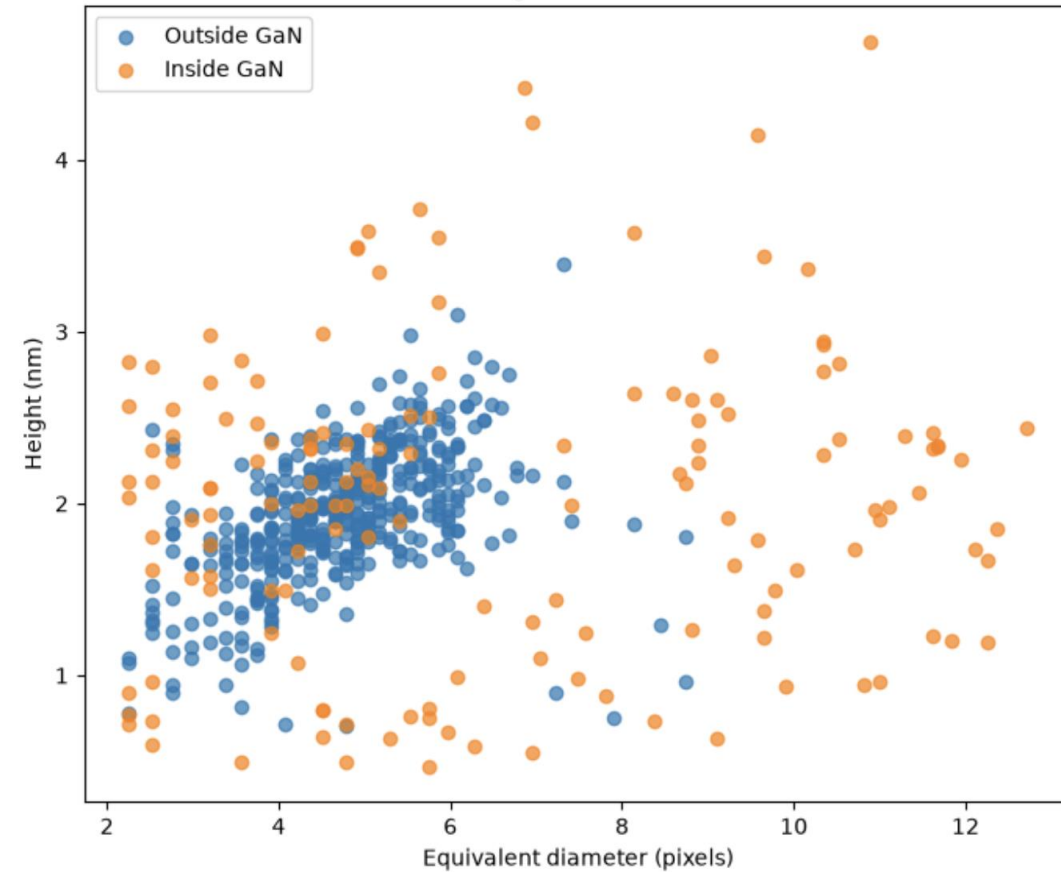
Error bars for histograms + error relative to the ground set



QD width distribution

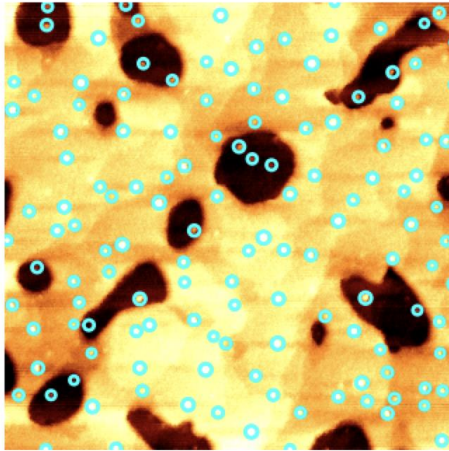


QD height versus width

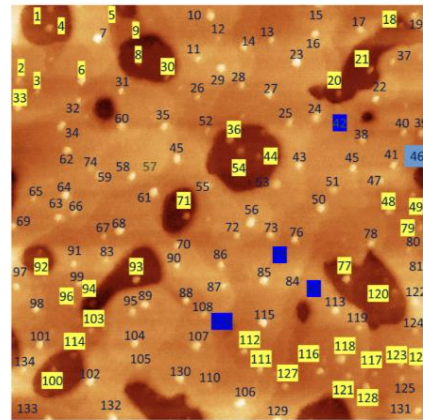


Ground Truth Dataset

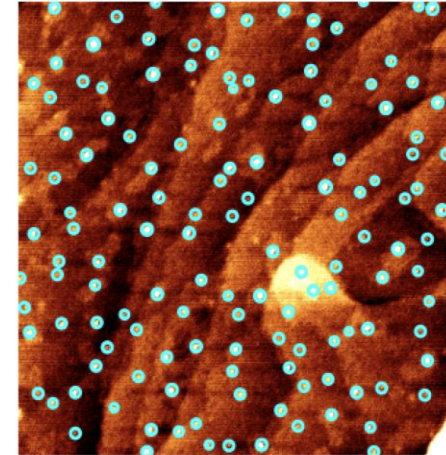
AFM image | QDs ≥ 6 Å: 134



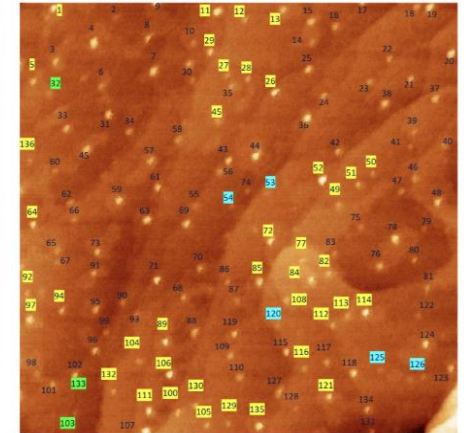
InGaN 2 nm



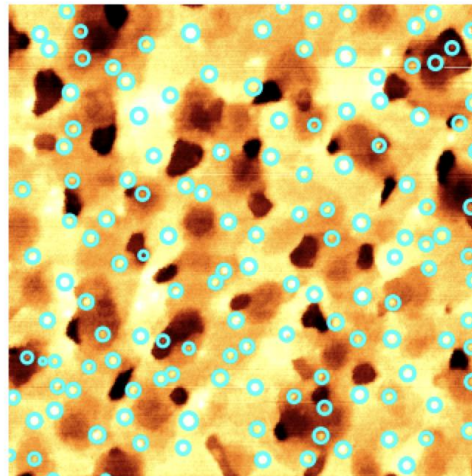
AFM image | QDs ≥ 6 Å: 138



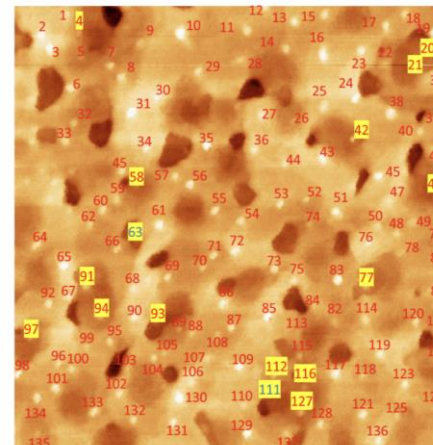
InGaN 1 nm



AFM image | QDs ≥ 6 Å: 138



InGaN 3 nm



Noise thresholds

At least 2 radii between QDs
because these cannot be too close
– otherwise they are a single object

How to count the dots in a
constellation? Prominence
criterion so that fake qds don't
cancel the real ones

Selecting QDs method?
Statistical relevance of the
dots

F1 SCORES